

Q1:-  $M(x, y, z) = 6x - y^2 + xz + 60$

$$g(x, y, z) = x^2 + y^2 + z^2 = 36 \Rightarrow x^2 + y^2 + z^2 - 36 = 0 \rightarrow (i)$$

$$L(x, y, z, \lambda) = 6x - y^2 + xz + 60 - \lambda(x^2 + y^2 + z^2 - 36)$$

$$\frac{\partial L}{\partial x} = 6 + z - 2\lambda x$$

$$\frac{\partial L}{\partial y} = -2y - 2\lambda y$$

$$\frac{\partial L}{\partial z} = x - 2\lambda z$$

$$\frac{\partial L}{\partial \lambda} = -x^2 - y^2 - z^2 + 36$$

$$6 + z - 2\lambda x = 0$$

$$-2y - 2\lambda y = 0$$

$$x - 2\lambda z = 0$$

$$-x^2 - y^2 - z^2 + 36 = 0$$

$$6 + z = 2\lambda x$$

$$-2y - 2\lambda y = 0 \quad -2y(1 + \lambda) = 0$$

$$-2y = 2\lambda y \Rightarrow \lambda = -1 \text{ or } y = 0$$

$$x = 2\lambda z$$

Case 1 :  $\lambda = -1$

$$6 + z = 2\lambda x$$

$$6 + z = -2x \rightarrow (a)$$

$$x = 2\lambda z$$

$$x = -2z$$

Substituting  $x$  in (a)

$$6 + z = -2(-2z)$$

$$z = 2 \quad \& \quad x = -4$$

Substituting  $z$  &  $x$  in (i)

$$(-4)^2 + y^2 + (2)^2 - 36 = 0$$

$$16 + y^2 + 4 - 36 = 0$$

$$y^2 - 16 = 0$$

$$y^2 = 16$$

$$y = \pm 4$$

Case 2:  $y = 0$

$$6 + z = 2x\lambda \rightarrow (b)$$

$$x = 2z\lambda$$

$$\lambda = \frac{x}{2z}$$

Substituting  $\lambda$  in (b)

$$6 + z = 2x \left( \frac{x}{2z} \right)$$

$$6 + z = \frac{x^2}{z}$$

$$6z + z^2 = x^2 \rightarrow (c)$$

Substituting  $x$  &  $y$  in (i)

$$(6z + z^2) + (0)^2 + z^2 = 36$$

$$6z + 2z^2 - 36 = 0$$

$$2z^2 + 6z - 36 = 0$$

$$z = -6 \text{ or } z = 3$$

If  $z = -6$  then eq (c) becomes  
 $x^2 = 0 \Rightarrow x = 0$

If  $z=3$  eq (c) becomes

$$x^2 = 27$$

$$x = \pm 3\sqrt{3}$$

So we have the pts  $(\pm 3\sqrt{3}, 0, 3), (0, 0, -6), (-4, \pm 4, 2)$

$$M(3\sqrt{3}, 0, 3) = 27\sqrt{3} + 60 \approx 106.8$$

$$M(-3\sqrt{3}, 0, 3) = 60 - 27\sqrt{3} \approx 13.2$$

$$M(0, 0, -6) = 60$$

$$M(-4, 4, 2) = 12$$

$$M(-4, -4, 2) = 12$$

Therefore the weakest field is at  $(-4, \pm 4, 2)$   
Q2 (a) Given that  $T$  is inversely proportional to the distance from origin

$$T = \frac{k}{\sqrt{x^2 + y^2 + z^2}} \rightarrow (i)$$

Given that temp at  $(1, 2, 2)$  is  $120^\circ$

$$120 = \frac{k}{\sqrt{1^2 + 2^2 + 2^2}}$$

$$120 = \frac{k}{\sqrt{9}}$$

$$120 = \frac{k}{3}$$

$$\boxed{360 = k}$$

$$\text{So temp} = T = \frac{360}{\sqrt{x^2 + y^2 + z^2}}$$

(b) Rate of change of  $T$  at  $(1, 2, 2)$  in the direction towards the pt  $(2, 1, 3)$



is  $D_u T(1,2,2)$

$$D_u f(x,y,z) = \nabla f(x,y,z) \cdot \vec{u}$$

$\vec{u}$  is the unit vector pointing towards  $(2,1,3)$  from  $(1,2,2)$   
 vector along unit vector  $\vec{u}$  is  $\langle 2-1, 1-2, 3-2 \rangle = \langle 1, -1, 1 \rangle$

$$\vec{u} = \left\langle \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$$

$$T_x = - \frac{360x}{(x^2+y^2+z^2)^{3/2}}$$

$$T_y = - \frac{360y}{(x^2+y^2+z^2)^{3/2}}$$

$$T_z = - \frac{360z}{(x^2+y^2+z^2)^{3/2}}$$

$$\nabla T(x,y,z) = - \frac{360x}{(x^2+y^2+z^2)^{3/2}}, - \frac{360y}{(x^2+y^2+z^2)^{3/2}}, - \frac{360z}{(x^2+y^2+z^2)^{3/2}}$$

$$\nabla T(1,2,2) = - \frac{360}{27}, - \frac{720}{27}, - \frac{720}{27}$$

$$\nabla T(1,2,2) = - \frac{40}{3}, - \frac{80}{3}, - \frac{80}{3}$$

$$D_u T(1,2,2) = - \frac{40}{3}, - \frac{80}{3}, - \frac{80}{3} \cdot \frac{1}{\sqrt{3}}, - \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$$

$$D_u T(1,2,2) = - \frac{40}{3\sqrt{3}} + \frac{80}{3\sqrt{3}} - \frac{80}{3\sqrt{3}} = - \frac{40}{3\sqrt{3}}$$

Rate of change of  $T$  at  $(1,2,2)$  in the direction towards the  
 pt  $(2,1,3)$  is  $-\frac{40}{3\sqrt{3}}$

$$\begin{aligned}
 \text{Sol:} \quad \lim_{(x,y) \rightarrow (0,2)} \frac{\sin(xy^2)}{xy} &= \lim_{(x,y) \rightarrow (0,2)} \frac{\sin(xy^2)}{xy^2} \cdot y. \\
 &= \lim_{(x,y) \rightarrow (0,2)} \frac{\sin(xy^2)}{xy^2} \cdot \lim_{(x,y) \rightarrow (0,2)} y. \\
 &= \lim_{(x,y) \rightarrow (0,2)} \frac{\sin(xy^2)}{xy^2} \cdot 2.
 \end{aligned}$$

$$\Rightarrow \lim_{(x,y) \rightarrow (0,2)} \frac{\sin(xy^2)}{xy} = 2 \cdot \lim_{(x,y) \rightarrow (0,2)} \frac{\sin(xy^2)}{xy^2} \quad \text{--- (*)}$$

$$\text{put } xy^2 = \theta.$$

$$\text{As } (x,y) \rightarrow (0,2), \text{ then } \theta = xy^2 \rightarrow (0)(2)^2 = 0.$$

then (\*) becomes.

$$\lim_{(x,y) \rightarrow (0,2)} \frac{\sin(xy^2)}{xy} = 2 \cdot \lim_{\theta \rightarrow 0} \frac{\sin(\theta)}{\theta}.$$

$$= 2(1)$$

$$\left( \because \lim_{\theta \rightarrow 0} \frac{\sin(\theta)}{\theta} = 1 \right)$$

$$\Rightarrow \boxed{\lim_{(x,y) \rightarrow (0,2)} \frac{\sin(xy^2)}{xy} = 2}$$

given  $f(x, y) = xy^2 + 5$ ,  $D = \{(x, y) \mid x \geq 0, y \geq 0, x^2 + y^2 \leq 3\}$

$$f_x(x, y) = y^2, f_y(x, y) = 2xy$$

for critical points,  $f_x(x, y) = 0$ ,  $f_y(x, y) = 0$

$$\Rightarrow y^2 = 0, 2xy = 0$$

$$\Rightarrow y = 0, x = 0$$

$\Rightarrow (x, y) = (0, 0)$  is the critical point

critical point lies in domain  $D$

on the boundary  $x = 0$ ,  $f(x, y) = 5$

on the boundary  $y = 0$ ,  $f(x, y) = 5$

on the boundary  $x^2 + y^2 = 3$ ,  $f(x, y) = xy^2 + 5$

$$f(x) = x(3 - x^2) + 5$$

$$f(x) = 3x - x^3 + 5$$

$$f'(x) = 3 - 3x^2$$

for extrema,  $f'(x) = 0$

$$3 - 3x^2 = 0$$

$$\Rightarrow x^2 = 1$$

$$\Rightarrow x = 1$$

$$\Rightarrow x=1$$

$$x=1, x^2+y^2=3$$

$$\Rightarrow 1^2+y^2=3$$

$$\Rightarrow y=\sqrt{2}$$

$(x,y)=(1,\sqrt{2})$  is a candidate point

$$f(1,\sqrt{2})=(1*\sqrt{2}^2)+5$$

$$f(1,\sqrt{2})=2+5$$

$$f(1,\sqrt{2})=7$$

absolute maximum value =7 ,absolute minimum value=5

$$f(x,y) = (\ln(y+1) + \sin(y/x)) / \sqrt{x^2 + y^2 - 4}$$

Here first of all notice that we want the denominator non zero for the function to be defined

Also as we want positive numbers inside sqrt hence domain of function  $f(x,y)$  is given such that :

$$x^2 + y^2 - 4 > 0$$

$$\Rightarrow x^2 + y^2 > 4$$

Also as  $\ln$  of positive numbers is defined, so we want

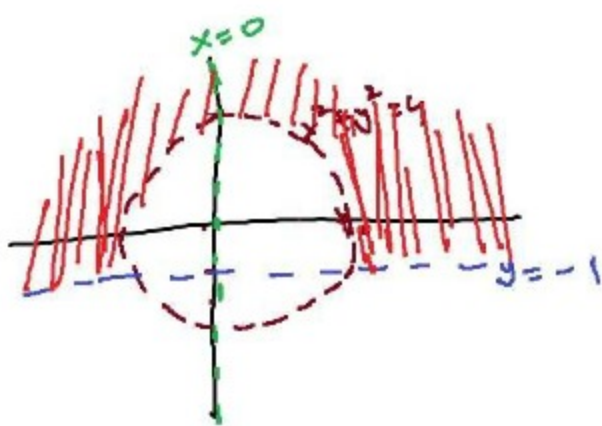
$$y + 1 > 0 \Rightarrow y > -1$$

And  $\sin(y/x)$  is defined only when  $x$  is not equal to 0

Hence the domain of function  $f(x,y)$  is defined as :

$$\{(x,y) : x \text{ not equal } 0, y > -1, x^2 + y^2 > 4\}$$

The sketch of the domain is in following figure with red shaded portion is the required region of domain:



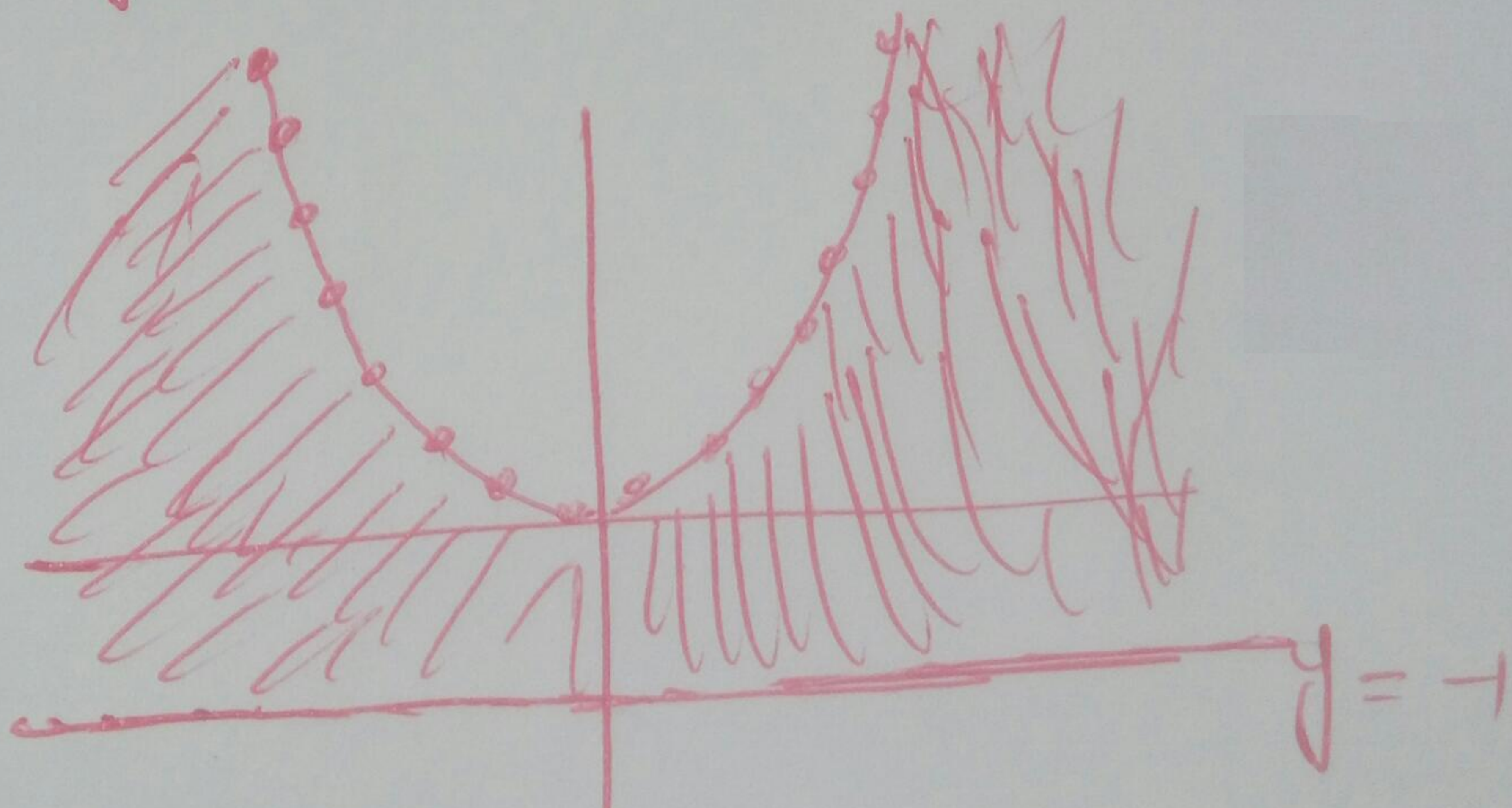
unbounded and open



Q4 (i)  $f(x, y) = \sqrt{y+1} + \ln(x^2 - y)$

$$y+1 \geq 0 \quad y \geq -1$$

$$x^2 - y > 0 \quad x^2 > y$$



neither open nor close